

TUTORIAL 3: APPLICATIONS OF DIFFERENTIATION

1. Use differentials to estimate the following expressions:
 - a) $(8.01)^{\frac{2}{3}} - (8.01)^{\frac{1}{3}} + 7$
 - b) $\sqrt{255.99} + \frac{1}{\sqrt[4]{255.99}}$
 - c) $\ln(0.99) + 3(0.99)^5$
 - d) $\sqrt{80.8} - \frac{4}{\sqrt[4]{80.8}}$
2. Given $f(x) = x^2\sqrt{1+x^3}$. Estimate the value of $f(2.02)$.
3. Given $f(x) = \frac{5}{x^2} + \ln(x-2)$. Estimate the value of $f(3.01)$.
4. At the depth of x cm, the volume of water in a container is given by
$$V = \frac{\pi x^2}{3}(12-x).$$
Find the volume of water that must be poured into the container so that the depth changes from 6 cm to 6.2 cm.
5. An empty container is filled with liquid. The height of the liquid in the container is x cm and its volume is V cm³, where $V = x(5 + 2x + x^2)$. If x increases at the rate of 0.35 cm per second, find the rate of change in the volume when x is 1 cm.
6. The radius of a sphere is increasing at the constant rate of 2 mm per sec. If the initial radius is 50 mm, find the radius of sphere after 15 seconds.
7. A conical water tank filled with water has a radius of r cm at the top and h cm high.
 - a) If the height of the water is twice the radius, show that $V = \frac{2}{3}\pi r^3$.
 - b) Water flows out of the tank at a rate of 15 cm³s⁻¹. How fast is the radius of the surface of the water changing when the height of the water is 4 cm?
8. Sketch the graph of $f(x) = 27 - (x-2)^3$.
9. Sketch the graph of $f(x) = \frac{2x-1}{x-3}$.
10. An empty container is filled with liquid. The height of the liquid in the container is x cm and its volume is V cm³ where $V = \frac{1}{2}\pi x(5-x)$. Find the value of x so that the volume of the container is maximum. What is the maximum volume?

Answer:

1. a) 9.0025 b) 16.2497 c) 2.84 d) 7.6547
2. 12.4
3. 0.5619
4. $2.4\pi \text{ cm}^3$
5. $4.2 \text{ cm}^3\text{s}^{-1}$
6. 80 mm
7. a) DIY b) $-\frac{15}{8}\pi$
8. DIY
9. DIY
10. $x = 2.5\text{cm}$ $V = 3.125\pi \text{ cm}^3$